ThmSmith Stage -1 proposal: `pid_network_exposure` / v1

Sharp ADE bounds from shared local-network completions under misspecified exposure maps

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Locked parameters. This is a partial-identification anchor, not a panel pool-mode. The locked tuple is

$$(\texttt{qid}, \texttt{specialization}, P, \theta, \text{identifying restrictions}, \text{bounding functional})$$
$$= (\texttt{pid\_network\_exposure}, \texttt{v1}, P_{I,A,S,Y;\pi,\mathcal{L}},\ \theta_{\text{ADE}}(s),\ \mathcal{R}_{\text{local graph+hidden completion}},\ [\underline{\theta}_{\text{joint}}(P), \overline{\theta}_{\text{joint}}(P)])$$

Here I indexes a uniformly sampled unit from a finite population, $A = (A_1, \ldots, A_n) \in \{0,1\}^n$ is assigned by a known randomized design π, $\mathcal{L}$ is the locally observed undirected graph, $S_i = s_i(A, \mathcal{L})$ is the researcher-specified local exposure, and $Y_i \in [0,1]$ is the observed outcome. The target $\theta_{\text{ADE}}(s)$ is the average direct treatment effect comparing $A_i = 1$ to $A_i = 0$ while holding the local exposure value $S_i = s$ fixed and averaging over the latent hidden-exposure state induced by the true graph completion. The restrictions $\mathcal{R}_{\text{local graph+hidden completion}}$ are finite support, consistency, bounded outcomes, randomized assignment, local exposure positivity, a completion class $\mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ of undirected true graphs that extend $\mathcal{L}$ by hidden candidate edges $\mathcal{C}$ subject to degree caps K_i, and exposure sufficiency for the pair $(S_i, R_i(G, A))$, where $R_i(G, A)$ is the hidden-neighbor exposure generated by the same graph completion G for all units. The bounding functional is the sharp finite linear-programming projection over one shared graph completion and bounded potential outcomes, not the product of independent unitwise exposure-propensity sensitivity sets.

Abstract

Network-interference analyses often specify a low-dimensional exposure mapping, but in many applications only the local neighborhood of each unit is known. Recent work gives sharp bounds for misspecified exposure mappings by bounding unitwise exposure propensities, and separate work studies bias or robust estimation when candidate networks are misspecified. This proposal asks whether sharp partial identification of an average direct treatment effect changes when hidden exposure errors must arise from one shared graph completion rather than from independent unitwise sensitivity perturbations. The routine theorem is a finite sharpness statement: the identified set is the projection of a graph-completion linear program. The flagship conjecture is that the usual cellwise propensity-ratio relaxation can be strictly nonsharp because one hidden edge simultaneously constrains the exposure labels of both endpoints. Resolving the conjecture would turn "local network knowledge" into testable compatibility inequalities for network exposure sensitivity analysis.

1 Introduction

Causal effects under interference depend on how treatments assigned to other units enter each unit's potential outcome. Exposure mappings make this problem tractable by compressing a high-dimensional assignment vector into a local summary, but the mapping is rarely known. The problem is sharper when the analyst knows only local network ties: a hidden tie is not merely a nuisance for one unit's exposure label, because the same edge also constrains the exposure label of the other endpoint.

The main proposed finding is a partial-identification result for average direct effects under locally known interference structure. Theorem 1 states the finite sharp identified set as a linear-programming projection over graph completions and bounded potential outcomes. Conjecture 1 states the novelty kernel: the standard cellwise exposure-sensitivity relaxation is generally too wide because it lets each unit choose an exposure-error law that no single hidden graph can realize jointly.

What is new is the shared-completion constraint. Egami (2020) treats spillover effects when the relevant network is unobserved, Schröder, Oprescu, Feuerriegel, and Kallus (2026) bound direct and spillover effects under misspecified exposure mappings through exposure-propensity ratio bounds, and Weinstein and Nevo (2026) study bias and robust estimation under candidate network misspecification. None isolates the finite partial-identification phenomenon that hidden local edges create cross-unit compatibility inequalities. This proposal is not another estimator and not a weaker version of known-exposure mapping theory; it asks for the sharp gap between a joint graph-completion identified set and the product relaxation obtained by forgetting that hidden exposures share edges.

The proposal sits in the partial-identification and sensitivity-analysis cluster. Its closest ancestors are Manski-style bounded-outcome identification, exposure-mapping interference estimands, and recent sensitivity analyses for misspecified exposure maps or misspecified networks. It deliberately avoids the panel / linear-projection cluster: the target is a finite-population average direct effect under interference, and the object to be characterized is a sharp identified interval over feasible graph completions.

2 Related work

Manski (1990) is the bounded-outcome partial-identification anchor: the present proposal uses the same logic that unobserved potential outcomes define intervals rather than point estimates. Hudgens and Halloran (2008) define direct and indirect effects under interference in the partial-interference setting, fixing the terminology for direct effects. Aronow and Samii (2017) introduce the exposure-mapping framework for general interference and show how randomized designs identify exposure-specific potential-outcome means when the mapping is correct. Forastiere, Airoldi, and Mealli (2021) extend network direct and interference effects to observational network studies with generalized propensity scores, illustrating how local exposures become estimands in applied network data.

Sävje, Aronow, and Hudgens (2021) show that a marginalized expected average treatment effect can be estimated under unknown limited interference, which is a useful contrast because this proposal targets a local-exposure direct effect that generally still needs network information. Leung (2022) replaces exact neighborhood interference by approximate neighborhood interference and establishes asymptotic inference for exposure effects; the present proposal instead keeps a finite partial-identification target and asks which graph completions are jointly feasible. Sävje (2024) separates exposure mappings as definitions from exposure mappings as structural restrictions, motivating sensitivity analyses when the mapping is not fully correct. The discussion by Leung (2024) emphasizes that once exposure mappings no longer impose the full structure, alternative restrictions are needed for inference; the shared graph-completion class is one such finite restriction.

Egami (2020) is the closest unobserved-network sensitivity comparator: it estimates network-specific spillover effects when one relevant network is unobserved, but it does not characterize sharp ADE bounds over a shared local graph-completion class. Bhattacharya, Malinsky, and Shpitser (2020) combines structure-learning and interference ideas for causal inference when network dependence is uncertain, whereas this proposal fixes a local completion class and asks for its finite identified set. Li, Sussman, and Kolaczyk (2021) studies how noisy network measurements affect bias and variance of network-interference estimators under a four-level exposure model; it does not give partial-identification bounds or shared-completion compatibility certificates. Schröder, Oprescu, Feuerriegel, and Kallus (2026) is the closest published partial-identification comparator: it gives sharp bounds for direct and spillover effects under exposure mapping misspecification through propensity-ratio sensitivity bounds, but it assumes a fully known network and does not impose shared hidden-edge compatibility. Weinstein and Nevo (2026) is the closest network-misspecification comparator: it bounds bias from using a wrong network and constructs an estimator robust to a list of candidate networks, but it does not derive sharp average-direct-effect bounds over a local graph-completion class. Yu, Airoldi, Borgs, and Chayes (2022) studies total treatment effects with unknown network structure, showing that some global targets can be handled without network measurement; the local direct-effect target here is precisely a case where that escape route is unavailable. Belloni, Fang, and Volfovsky (2022) adaptively estimates unknown interference radius on a known network, whereas this proposal treats hidden graph completion as the partial-identification object. Gao, Harshaw, Sävje, and Wang (2026) proves that broad specification tests for exposure mapping models are impossible without additional restrictions, strengthening the case for reporting sharp bounds under explicit local-knowledge restrictions.

The in-repository catalogue does not already contain this kernel. The relevant entry in `doc/API.md` is the ExactID / PartialID catalogue note, which records that the PartialID catalogue is a seed area for Manski-family bounds, IV bounds, sensitivity analysis, shape restrictions, and missing-data bounds rather than a populated network-exposure theorem. The closest banked partial-ID proposal is `pid_dynamic_iv_compliance_v2`: assignment-memory mixture splitting for dynamic IV bounds under imperfect compliance. That proposal also uses finite latent-response sharpness, but its kernel is compliance-history memory, not shared graph-completion compatibility. The failed `pid_late_bounded_defiers` proposal concerns bounded-defier complier-LATE envelopes and is likewise outside network interference.

3 Formal setup

There are $n < \infty$ units. The assignment vector is $A \in \{0,1\}^n$, drawn from a known design $\pi(a) > 0$ on its support. The analyst observes a local undirected graph $\mathcal{L}$ and a candidate hidden-edge set $\mathcal{C} \subseteq \{\{i,j\} : i < j\}$ disjoint from $\mathcal{L}$. A feasible true graph $G \in \mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ contains all edges in $\mathcal{L}$, may contain hidden edges from $\mathcal{C}$, and satisfies $\sum_{j:\{i,j\}\in G\setminus\mathcal{L}} 1 \le K_i$ for each unit i.

For $e = \{i,j\} \in \mathcal{C}$, let $x_e(G) = \mathbf{1}\{e \in G\}$, and let $B_{ie} = \mathbf{1}\{i \in e\}$ be the hidden-edge endpoint incidence matrix. Degree-cap feasibility is equivalently the capacitated incidence system

$$x_e \in \{0,1\}, \qquad \sum_{e\in\mathcal{C}} B_{ie} x_e \le K_i \quad \text{for every } i.$$

For any assignment a, the hidden-exposure requirement that unit i has at least one treated hidden neighbor is

$$\sum_{e=\{i,j\}\in\mathcal{C}} x_e a_j \ge z_i(a),$$

where $z_i(a) \in \{0,1\}$ is the exposure-event variable used in the certificate formulation; if $z_i(a) = 0$, no lower requirement is imposed. A strict cellwise endpoint is graph-realizable only if one binary vector x satisfies all endpoint-incidence, degree-cap, and exposure-event requirements that the endpoint construction demands.

The researcher-specified local exposure is $S_i = s_i(A, \mathcal{L}) \in \mathcal{S}$. For a graph completion G, the hidden exposure is

$$R_i(G, A) = \mathbf{1}\{\exists j : \{i,j\} \in G \setminus \mathcal{L},\ A_j = 1\} \in \{0,1\}.$$

The structural exposure state is $H_i(G, A) = (S_i, R_i(G, A))$. Potential outcomes are $Y_i(a_i, s, r) \in [0,1]$, and consistency requires $Y_i = Y_i(A_i, S_i, R_i(G, A))$ for the true completion G. The observable distribution is the finite-population law $P_{I,A,S,Y;\pi,\mathcal{L}}$ generated by a uniformly sampled unit I, the known assignment law π, the observed local exposure S_I, and the outcome Y_I.

For a local exposure value s with positive design probability, define

$$q_{i,s,r}(G) = \mathbb{P}_\pi(R_i(G, A) = r \mid S_i = s, A_i = 0)$$

and the average direct effect

$$\theta_{\mathrm{ADE}}(s; G, Y(\cdot)) = \frac{1}{n_s(G)} \sum_{i:\mathbb{P}_\pi(S_i=s,A_i=0)>0} \sum_{r\in\{0,1\}} q_{i,s,r}(G)\{Y_i(1,s,r) - Y_i(0,s,r)\},$$

where $n_s(G)$ is the number of units in the displayed sum. The identified set $\Theta_I(P)$ is the set of all values of $\theta_{\mathrm{ADE}}(s; G, Y(\cdot))$ over feasible graph completions and bounded potential outcomes satisfying consistency and matching the observed distribution P. The bounding functionals are $\underline{\theta}_{\mathrm{joint}}(P) = \inf \Theta_I(P)$ and $\overline{\theta}_{\mathrm{joint}}(P) = \sup \Theta_I(P)$.

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> $$\begin{aligned} &(\texttt{qid}, \texttt{specialization}, P, \theta, \text{identifying restrictions}, \text{bounding functional}) \\ &= (\texttt{pid\_network\_exposure}, \texttt{v1}, P_{I,A,S,Y;\pi,\mathcal{L}},\ \theta_{\mathrm{ADE}}(s),\ \mathcal{R}_{\text{local graph+hidden completion}},\ [\underline{\theta}_{\mathrm{joint}}(P), \overline{\theta}_{\mathrm{joint}}(P)]). \end{aligned}$$
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> Here I indexes a uniformly sampled unit from a finite population, $A = (A_1, \dots, A_n) \in \{0,1\}^n$ is assigned by a known randomized design π, $\mathcal{L}$ is the locally observed undirected graph, $S_i = s_i(A, \mathcal{L})$ is the researcher-specified local exposure, and $Y_i \in [0,1]$ is the observed outcome. The target $\theta_{\mathrm{ADE}}(s)$ is the average direct treatment effect comparing $A_i = 1$ to $A_i = 0$ while holding the local exposure value $S_i = s$ fixed and averaging over the latent hidden-exposure state induced by the true graph completion. The restrictions $\mathcal{R}_{\text{local graph+hidden completion}}$ are finite support, consistency, bounded outcomes, randomized assignment, local exposure positivity, a completion class $\mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ of undirected true graphs that extend $\mathcal{L}$ by hidden candidate edges $\mathcal{C}$ subject to degree caps K_i, and exposure sufficiency for the pair $(S_i, R_i(G, A))$, where $R_i(G, A)$ is the hidden-neighbor exposure generated by the same graph completion G for all units. The bounding functional is the sharp finite linear-programming projection over one shared graph completion and bounded potential outcomes, not the product of independent unit-wise exposure-propensity sensitivity sets.

4 Assumptions

Assumption 1 (Finite randomized local-network experiment)**.** The population has $n < \infty$ units, $A \in \{0,1\}^n$ is drawn from a known design π, the local graph $\mathcal{L}$, candidate hidden-edge set $\mathcal{C}$, degree caps K_i, local exposure functions s_i, and outcome Y_i are observed as in Section 6.

Assumption 2 (Bounded potential outcomes)**.** For every unit i, treatment $a \in \{0,1\}$, local exposure $s \in \mathcal{S}$, and hidden exposure $r \in \{0,1\}$, $0 \le Y_i(a, s, r) \le 1$.

Assumption 3 (Shared hidden graph completion)**.** There exists one undirected graph $G \in \mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ that determines $R_i(G, A)$ for all units and all assignments. Hidden edge choices are not unit-specific nuisance variables.

Assumption 4 (Exposure sufficiency and consistency)**.** For the true graph completion G, if $A_i = a$, $S_i = s$, and $R_i(G, A) = r$, then the observed outcome satisfies $Y_i = Y_i(a, s, r)$. Potential outcomes depend on other units' assignments only through $(S_i, R_i(G, A))$.

Assumption 5 (Local positivity)**.** For the target exposure value s, every unit included in $\theta_{\text{ADE}}(s)$ has positive design probability for $A_i = 0, S_i = s$ and $A_i = 1, S_i = s$. Every observed cell used to match P has positive probability under π.

Assumption 6 (Observable-law matching)**.** A feasible pair $(G, Y(\cdot))$ must reproduce the observed conditional distribution of Y_I given (I, A_I, S_I) under the known design and local graph. Equivalently, for each observed cell, the average over assignments and units induced by $(G, Y(\cdot))$ equals the corresponding component of $P_{I,A,S,Y;\pi,\mathcal{L}}$.

Assumption 7 (Cellwise relaxation comparator)**.** The cellwise comparator $\Theta_{\text{cell}}(P)$ is a separate finite relaxation. It keeps Assumptions 1, 2, and 5, but replaces the shared completion G and the graph-based consistency equation in Assumptions 3–4 by independent latent exposure variables $Z_i(a) \in \{0, 1\}$ with kernels $Q_i(z \mid a_i, s_i(a, \mathcal{L}))$. These kernels must have the same unitwise support and marginal exposure-propensity-ratio ranges that some feasible completion in $\mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ could generate for unit i in isolation. Observed-law matching is imposed directly through the mixture equations

$$\mathbb{E}_{Z_i \sim Q_i(\cdot \mid A_i, S_i)}\{Y_i(A_i, S_i, Z_i) \mid I = i, A_i, S_i\} = \mathbb{E}_P(Y_I \mid I = i, A_i, S_i),$$

with $0 \leq Y_i(a, s, z) \leq 1$. No true graph G, no shared edge vector x, and no equation involving $R_i(G, A)$ is assumed inside $\Theta_{\text{cell}}(P)$.

5 Main results

Theorem 1 (Theorem 1: finite joint-completion LP sharpness)**.** *Under Assumptions 1–6, the sharp identified set for $\theta_{\text{ADE}}(s)$ is*

$$\Theta_I(P) = \{\theta_{\text{ADE}}(s; G, Y(\cdot)) : G \in \mathfrak{G}(\mathcal{L}, \mathcal{C}, K),\ 0 \leq Y_i(a, s, r) \leq 1,\ (G, Y(\cdot)) \textit{ matches } P\}.$$

Because $\mathfrak{G}(\mathcal{L}, \mathcal{C}, K)$ is finite, the endpoints $\underline{\theta}_{\text{joint}}(P)$ and $\overline{\theta}_{\text{joint}}(P)$ are obtained by solving a finite collection of linear programs, one for each graph completion, and taking the minimum lower endpoint and maximum upper endpoint.

Why this is new and worth studying. The closest papers are Schröder, Oprescu, Feuerriegel, and Kallus (2026) and Weinstein and Nevo (2026). The former gives exposure-propensity sensitivity bounds without hidden graph-completion compatibility, while the latter studies bias and robust estimation across candidate networks rather than the sharp partial-identification projection for a direct effect. This theorem is routine finite-dimensional sharpness, but it fixes the correct object for the genuinely new compatibility conjecture and gives Stage 0 a precise bounding functional.

Conjecture 1 (Conjecture 1: shared-edge compatibility makes cellwise bounds nonsharp)**.** *(NOT YET PROVED.) Under Assumptions 1–6, and with $\Theta_{\mathrm{cell}}(P)$ defined by Assumption 7, there exists an open set of finite experiments with at least one hidden candidate edge $\{i, j\} \in \mathcal{C}$ such that*

$$\Theta_I(P) \subsetneq \Theta_{\mathrm{cell}}(P).$$

The strict containment can occur even when every unit-level hidden-exposure propensity allowed by the cellwise relaxation is generated by some feasible graph completion for that unit in isolation. The obstruction is joint: the same hidden edge must simultaneously determine $R_i(G, A)$ and $R_j(G, A)$, so endpoint assignments permitted by independent unitwise kernels need not be realizable by any single G.

Why this is new and worth studying. The closest published result is the sharp propensity-ratio framework of Schröder, Oprescu, Feuerriegel, and Kallus (2026); the epsilon-gap is that their sensitivity object is local to an exposure propensity, whereas this conjecture says the sharp set changes when exposure errors must be induced by one shared network. Egami (2020), Bhattacharya, Malinsky, and Shpitser (2020), and Li, Sussman, and Kolaczyk (2021) are closer on network uncertainty, but they target estimation, discovery, or bias correction rather than finite sharp ADE bounds under one shared completion. Weinstein and Nevo (2026) recognizes that the correctly specified network may not be unique, but it does not characterize the finite compatibility facets that distinguish a joint hidden graph from a product of unitwise errors. Proving or refuting the conjecture would tell practitioners when exposure-mapping sensitivity intervals are conservative because they forgot edge sharing, and it would supply concrete compatibility checks for locally measured networks.

Conjecture 2 (Conjecture 2: capacitated-incidence certificate characterization)**.** *(NOT YET PROVED.) Under Assumptions 1–6, and with* $\Theta_{\mathrm{cell}}(P)$

defined by Assumption 7, every strict gap between $\Theta_I(P)$ and $\Theta_{\text{cell}}(P)$ has a finite certificate supported on the capacitated incidence system linking hidden candidate edges to unit-assignment exposure events. After fixing an endpoint value in $\Theta_{\text{cell}}(P)$, let $\mathcal{D}$ be the finite set of demanded exposure events (i,a) for which the endpoint construction requires $z_i(a)=1$. Graph-realizability requires binary hidden-edge variables x_e satisfying

$$\sum_{e\in\mathcal{C}} B_{ie}x_e \le K_i \quad \text{for all } i, \qquad \sum_{e=\{i,j\}\in\mathcal{C}} x_e a_j \ge 1 \quad \text{for all } (i,a)\in\mathcal{D}.$$

The conjecture is that failure of this system, together with the observable-law equalities, is always witnessed by a violated finite certificate from the associated capacitated b-matching / set-covering polytope, including endpoint-incidence degree consumption and any odd-set or lifted cover inequalities needed for binary graph feasibility.

Why this is new and worth studying. No cited network-interference paper gives this finite compatibility certificate. Belloni, Fang, and Volfovsky (2022) search adaptively over local patterns on a known network, Egami (2020) develops sensitivity analysis for unobserved networks, and Yu, Airoldi, Borgs, and Chayes (2022) avoid local-network measurement for a total-effect target; none supplies a capacitated graph-completion certificate for a local direct-effect bound. If true, the conjecture would turn the proposed LP from an exhaustive enumeration into interpretable diagnostics: a bound is loose exactly because a specific collection of hidden-edge requirements cannot coexist under endpoint incidence and degree caps.

6 Sanity-check corollaries

Corollary 1 (Known-network collapse). *Under Assumptions 1–6, if $\mathcal{C}=\emptyset$ or $K_i=0$ for all i, then $\mathfrak{G}(\mathcal{L},\mathcal{C},K)=\{\mathcal{L}\}$, $R_i(G,A)=0$, and Theorem 1 reduces to the usual exposure-mapping finite-population direct-effect calculation under a known graph.*

Under Assumptions 3 and 4, the hidden exposure has only the value $r=0$, so the target collapses to the known-exposure contrast because the sum over r has one term.

Corollary 2 (No hidden-effect collapse). *If Assumptions 1–6 hold and $Y_i(a,s,0)=Y_i(a,s,1)$ for every i,a,s, then every feasible graph completion gives the same $\theta_{\text{ADE}}(s)$. The joint and cellwise identified sets coincide.*

Under the displayed equality, $R_i(G, A)$ drops out of every potential outcome, so changing G only relabels a nuisance state that has no effect on the direct-effect contrast.

7 Proof-sketch route

Theorem 1 reduces to finite support and the classical sharpness argument for bounded missing potential outcomes: condition on a graph completion, write each observed conditional mean as a linear equality in the bounded latent outcomes $Y_i(a, s, r)$, optimize the linear functional defining $\theta_{\mathrm{ADE}}(s)$, and take the union over finitely many completions. Conjecture 1 should be attacked first by a four-unit boundary example with one or two hidden candidate edges, enumerating all $2^{|\mathcal{C}|}$ completions and comparing the resulting endpoint to the product relaxation. Conjecture 2 is a finite integer/linear-programming duality question: project the capacitated incidence constraints $Bx \leq K$ and the assignment-specific covering constraints onto exposure-event requirements, then identify the b-matching, cover, or odd-set inequalities that separate infeasible cellwise endpoints. The two sanity corollaries are direct substitutions into the target formula and should be checked before any larger witness is trusted.

8 Honest scope flags

The LP sharpness theorem is labelled as a theorem because it is routine finite partial-identification algebra, not because it has been proved in this proposal. The strict nonsharpness and cut-certificate statements are conjectures and are the real novelty kernel. The proposal does not claim statistical inference, asymptotic coverage, or an efficient estimator; it is only about the population identified set. The exact minimal number of units needed for strict containment is intentionally left open. The most delicate modelling choice is the target's use of $q_{i,s,r}(G)$ conditional on $A_i = 0$; Stage 0 may instead prefer a symmetric reference distribution over R_i conditional only on $S_i = s$, but the shared-completion phenomenon should survive that convention.

9 Iteration trail

- v1 (cold-start) – initial draft proposing shared hidden-edge graph-completion bounds for locally known network exposures; prior verdict: none.

- v2 (revise) – added missing unobserved-network comparators, separated the cellwise relaxation from true-graph consistency, and replaced the cut display with a capacitated-incidence certificate statement; prior verdict: REVISE.

Bibliography

References

Aronow, P. M. and Samii, C. (2017). Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics* 11(4), 1912–1947.

Belloni, A., Fang, F., and Volfovsky, A. (2022). Neighborhood adaptive estimators for causal inference under network interference. arXiv:2212.03683.

Bhattacharya, R., Malinsky, D., and Shpitser, I. (2020). Causal inference under interference and network uncertainty. *Proceedings of the 35th Conference on Uncertainty in Artificial Intelligence*, PMLR 115, 1028–1038.

Egami, N. (2020). Spillover effects in the presence of unobserved networks. *Political Analysis* 29(3), 287–316.

Forastiere, L., Airoldi, E. M., and Mealli, F. (2021). Identification and estimation of treatment and interference effects in observational studies on networks. *Journal of the American Statistical Association* 116(534), 901–918.

Gao, C., Harshaw, C., Sävje, F., and Wang, Y. (2026). On the impossibility of specification testing of interference models based on exposure mappings. arXiv:2605.09726.

Hudgens, M. G. and Halloran, M. E. (2008). Toward causal inference with interference. *Journal of the American Statistical Association* 103(482), 832–842.

Leung, M. P. (2022). Causal inference under approximate neighborhood interference. *Econometrica* 90(1), 267–293.

Leung, M. P. (2024). Discussion of "Causal inference with misspecified exposure mappings: separating definitions and assumptions". *Biometrika* 111(1), 17–20.

Li, W., Sussman, D. L., and Kolaczyk, E. D. (2021). Causal inference under network interference with noise. arXiv:2105.04518.

Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review* 80(2), 319–323.

Sävje, F. (2024). Causal inference with misspecified exposure mappings: separating definitions and assumptions. *Biometrika* 111(1), 1–15.

Sävje, F., Aronow, P. M., and Hudgens, M. G. (2021). Average treatment effects in the presence of unknown interference. *The Annals of Statistics* 49(2), 673–701.

Schröder, M., Oprescu, M., Feuerriegel, S., and Kallus, N. (2026). Causal inference on networks under misspecified exposure mappings: A partial identification framework. arXiv:2602.03459.

Weinstein, B. and Nevo, D. (2026). Causal inference with misspecified network interference structure. *Biometrics* 82(1), ujag023.

Yu, C. L., Airoldi, E. M., Borgs, C., and Chayes, J. T. (2022). Estimating the total treatment effect in randomized experiments with unknown network structure. *Proceedings of the National Academy of Sciences* 119(44), e2208975119.